

Ride Booking Big Data Project

Complete Hive Command Documentation

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Hive Data Cleaning & Dashboard Workflow Documentation

Course: STQD6324 Data Management University: Universiti Kebangsaan Malaysia (UKM) Project: Ride Booking Analytics Dashboard Platform: Apache Hive • HDFS • Hadoop • R Shiny

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Part 1. Hive Data Cleaning Workflow

This section documents the complete data cleaning workflow performed in Apache Hive. The process includes dataset upload, table creation, data transformation, validation, duplicate checking, missing value handling, and exporting the cleaned dataset for downstream analytics.

1. Create HDFS Directory and Upload Dataset

Purpose

Create a project directory in HDFS and upload the booking dataset for further Hive processing.

Create HDFS Directory

```
# Create project directory in HDFS
```

```
hdfs dfs -mkdir /ride
```

Upload Dataset to HDFS

```
# Upload CSV file into HDFS
```

```
hdfs dfs -put Bookings.csv /ride/
```

Verify Uploaded File

```
# Verify uploaded dataset
```

```
hdfs dfs -ls /ride
```

Check Dataset Header

```
hdfs dfs -cat /ride/Bookings.csv | head -1
```

Display Column Names

```
hdfs dfs -cat /ride/Bookings.csv | head -1 | tr ',' '\n'
```

2. Create Hive Raw Table

Purpose

Create the raw Hive table to store the original booking dataset before any cleaning operations.

```
CREATE TABLE bookings_raw (  
...  
);
```

□□□□□□□□□□ SQL□

3. Load Dataset into Hive

Purpose

Load the CSV dataset stored in HDFS into the Hive raw table.

```
LOAD DATA INPATH '/ride/Bookings.csv'  
INTO TABLE bookings_raw;
```

4. Verify Raw Data

Purpose

Validate that the dataset has been successfully loaded into Hive.

Check Table Structure

```
DESCRIBE bookings_raw;
```

Check Total Records

```
SELECT COUNT(*)  
FROM bookings_raw;
```

Preview Dataset

```
SELECT *  
FROM bookings_raw  
LIMIT 10;
```

5. Data Cleaning and Transformation

Purpose

The following cleaning procedures are performed to improve data quality before analytics.

Remove duplicate records

Replace invalid "null" strings with actual NULL values

Convert numeric attributes into DOUBLE

Standardize data format

Generate a cleaned Hive table

```
DROP TABLE IF EXISTS bookings_clean;  
CREATE TABLE bookings_clean AS  
SELECT DISTINCT  
...  
FROM bookings_raw;
```

□□□□□□ SQL

6. Validate Cleaned Dataset

Purpose

Verify the newly created cleaned dataset.

```
SHOW TABLES;
DESCRIBE bookings_clean;
SELECT COUNT(*)
FROM bookings_clean;
SELECT *
FROM bookings_clean
LIMIT 20;
```

7. Check Missing Values

Purpose

Evaluate data completeness after cleaning.

```
SELECT
...
FROM bookings_clean;
```

8. Check Duplicate Records

Purpose

Ensure duplicate booking records have been successfully removed.

```
SELECT
Booking_ID,
COUNT(*) AS Duplicate_Count
FROM bookings_clean
GROUP BY Booking_ID
HAVING COUNT(*) > 1;
```

9. Export Cleaned Dataset to HDFS

Purpose

Export the cleaned dataset back into HDFS for downstream processing.

```
INSERT OVERWRITE DIRECTORY '/ride_output'
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
SELECT *
FROM bookings_clean;
```

10. Download Cleaned Dataset

Merge Output Files

```
hdfs dfs -getmerge /ride_output ~/Bookings_Cleaned_Hive.csv
```

Verify Output File

```
head ~/Bookings_Cleaned_Hive.csv
```

11. Store Final Dataset in HDFS

Create Storage Directory

```
hdfs dfs -mkdir -p /ride_final
```

Upload Final Dataset

```
hdfs dfs -put -f ~/Bookings_Cleaned_Hive.csv /ride_final/
```

Verify Uploaded Dataset

```
hdfs dfs -ls /ride_final
```

12. Final Validation

```
hdfs dfs -cat /ride_final/Bookings_Cleaned_Hive.csv | head
```

Part 2. Hive Dashboard Analysis Tables

The following Hive tables are created specifically for the Ride Booking Analytics Dashboard. Each table supports one or more visualizations in the R Shiny dashboard.

1. Create KPI Dashboard Table

Purpose

Generate overall business KPI metrics.

```
CREATE TABLE dashboard_kpi AS
```

...

2. Create Booking Trend Table

Purpose

Generate daily booking trends.

```
CREATE TABLE dashboard_trend AS
```

...

3. Create Vehicle Revenue Table

Purpose

Calculate revenue generated by each vehicle type.

```
CREATE TABLE dashboard_vehicle AS
```

...

4. Create Payment Method Table

Purpose

Summarize booking counts by payment method.

```
CREATE TABLE dashboard_payment AS
```

```
...
```

5. Create Cancellation Status Table

Purpose

Calculate booking status distribution.

```
CREATE TABLE dashboard_cancel AS
```

```
...
```

6. Create Customer Rating Table

Purpose

Extract customer ratings for dashboard visualization.

```
CREATE TABLE dashboard_rating AS
```

```
...
```

7. Create Detailed Records Table

Purpose

Store the complete cleaned dataset for dashboard display.

```
CREATE TABLE dashboard_detail AS
```

```
SELECT *
```

```
FROM bookings_clean;
```

8. Create Revenue by Vehicle Table

Purpose

Generate vehicle revenue statistics.

```
CREATE TABLE dashboard_revenue AS
```

```
...
```

9. Create Pickup Location Table

Purpose

Summarize booking counts by pickup location.

```
CREATE TABLE dashboard_pickup AS
```

```
...
```

10. Create Customer Analysis Table

Purpose

Calculate booking frequency and spending for each customer.

```
CREATE TABLE dashboard_customer AS
```

```
...
```

11. Verify Dashboard Tables

```
SHOW TABLES LIKE 'dashboard_%';
```

12. Preview Dashboard Data

```
SELECT *
```

```
FROM dashboard_trend
```

```
LIMIT 10;
```

Dashboard Tables Summary

Workflow Summary

- ☐ Dataset uploaded into HDFS
- ☐ Hive raw table created
- ☐ Dataset loaded into Hive
- ☐ Data cleaned and transformed
- ☐ Missing values processed
- ☐ Duplicate records removed
- ☐ Cleaned dataset validated
- ☐ Dashboard tables generated
- ☐ Dataset exported back to HDFS
- ☐ Ready for R Shiny Dashboard and Business Analytics